Personality & Emotion for AI Design

BCSAI - Spring 2025

Session 2 - Neurons and the Basics of Neural Networks

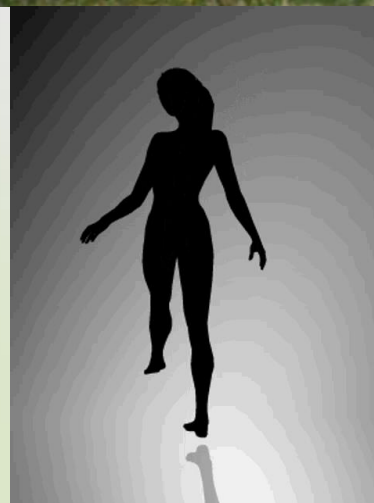
Many slides adapted from Lafferty (Yale), Weinberger (Cornell), Szameitat (Brunel)

Session 2 - Neurons and the Basics of Neural Networks

- Basic neuroscience: How biological neurons work
- Similarities between brain and AI,
- Similarities between biological and artificial neurons
- Models of artificial neurons
- The perceptron: A simple neural network model

The brain

When the brain receives any ambiguous or even cluttered sensory input, we tend to impose **consistent** interpretations, at least consciously. In recent decades it has become clear that unconsciously the brain **can consider many** interpretations of many **inputs**, some of them **inconsistent** with each other. Our brain's ability to extract consistency in an inconsistent-to-lifelike image underlies our love for and understanding of abstract art forms, such as Impressionism.



Fundamentals of cognitive neurscience: A beginner´s guide. Gage & Baars

The brain

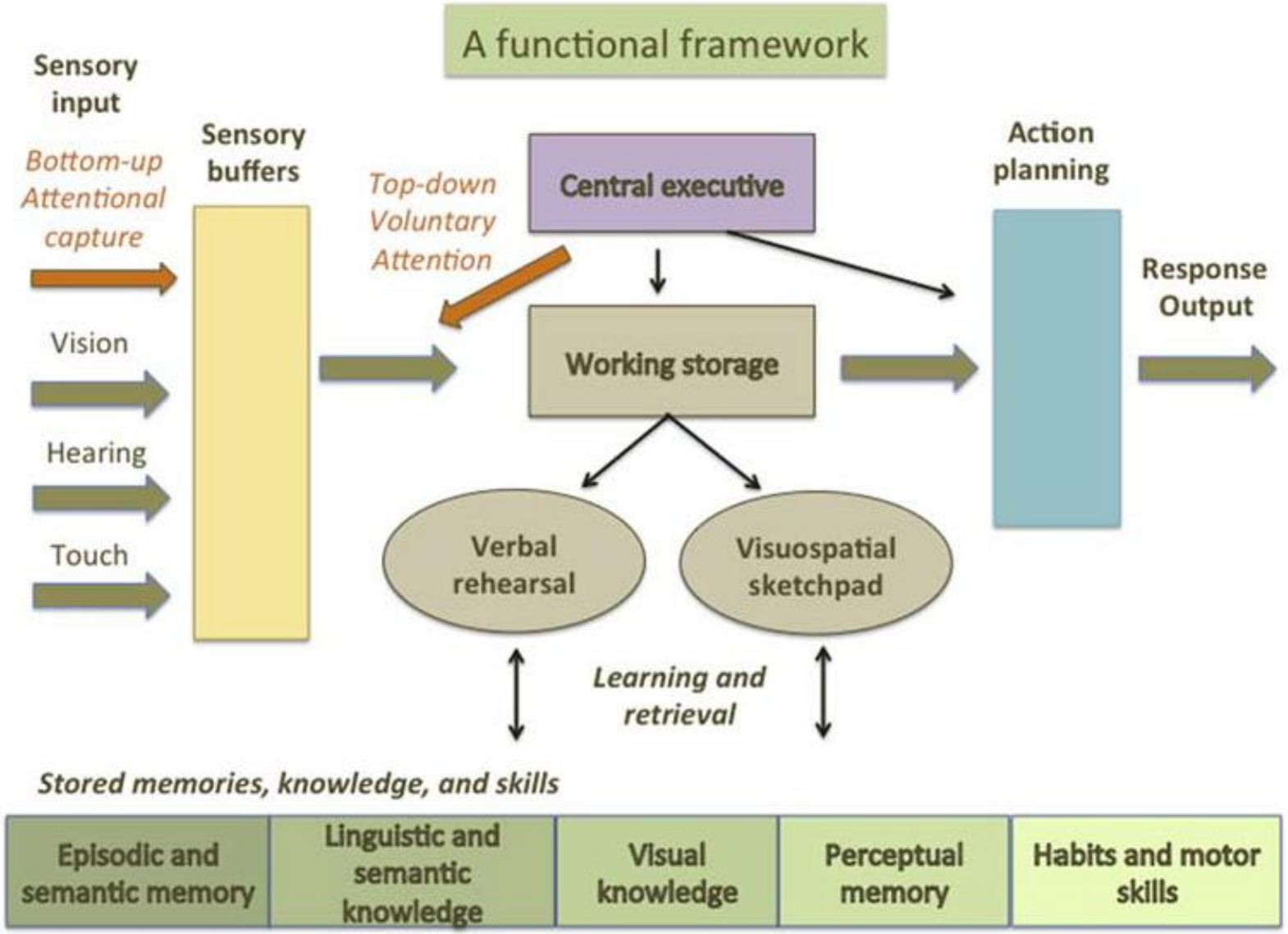
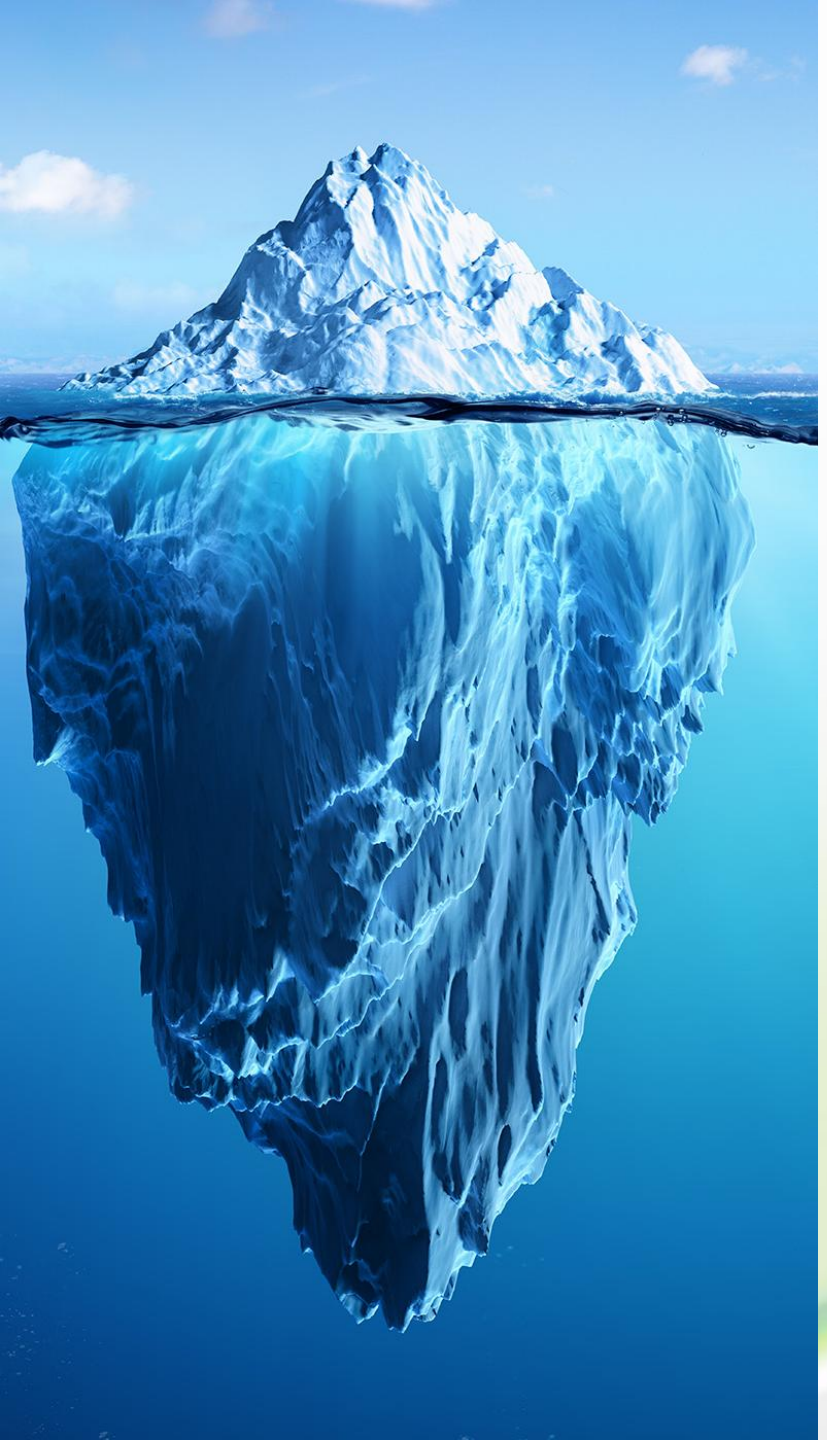


FIGURE 1.4 A functional framework for human cognition. Sensory inputs (vision, hearing, touch, left side of figure) enter the brain system through the sensory pathways. The executive and attentional processes governed by frontal lobe regions serve to focus attention, hold information temporarily in working storage (center of figure), and prepare it for long-term storage (green boxes at bottom of figure). Action plans and response outputs are shown (right side of figure). Note that green-shaded boxes grow lighter in color as the processes and information they represent are mediated by more and more unconscious processing. *Source: Baars & Gage.*

Fundamentals of cognitive neurscience: A beginner´s guide. Gage & Baars



The brain

More and more evidence is providing support for the notion that it is our **unconscious** processing that forms the overwhelming majority of brain functions. Our vast store of memories, **language knowledge**, **automaticities**, and **procedural learning** combine to form the largely unconscious but accessible if recalled storage of what we have learned and experienced throughout our life. Shown in the green boxes at the base of [previous] Fig. 1.4, this information is **retrievable**, in large part, and forms the bulk of the iceberg of our knowledge store. Conscious contents, at any given moment, form merely the tip of this massive iceberg.

Fundamentals of cognitive neurscience: A beginner´s guide. Gage & Baars

The brain and AI: Similarities

✔ Most “knowledge” is latent and inaccessible: Neural networks store vast amounts of information in **weights**, which:

- **Cannot be** directly **inspected** in a human-readable way
- Represent statistical patterns **learned** from **data**
- Only the small, explicit part that the model **outputs** during inference is **visible**.

✔ Computations are mostly **automatic**:

- Neural networks operate via automatic activation of layers
- Don’t “choose” which memories to activate
- **don’t have introspection** or access to their hidden states

The brain and AI: Differences

- ❌ No conscious/unconscious divide
- ❌ AI does not recall or retrieve memories
- ❌ No self-model, awareness nor introspection
- ❌ Narrow

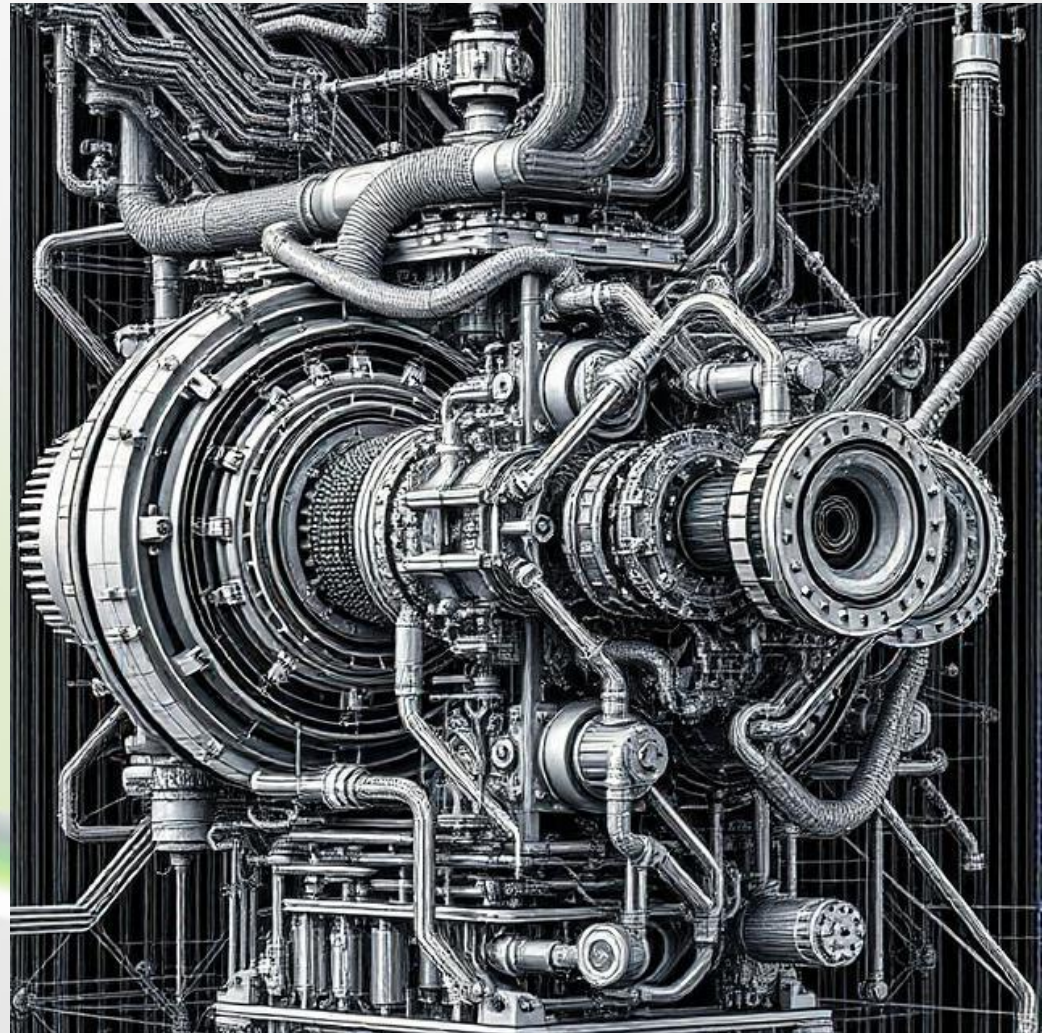
❓ Q: Can AI think outisde the box?

The brain and AI

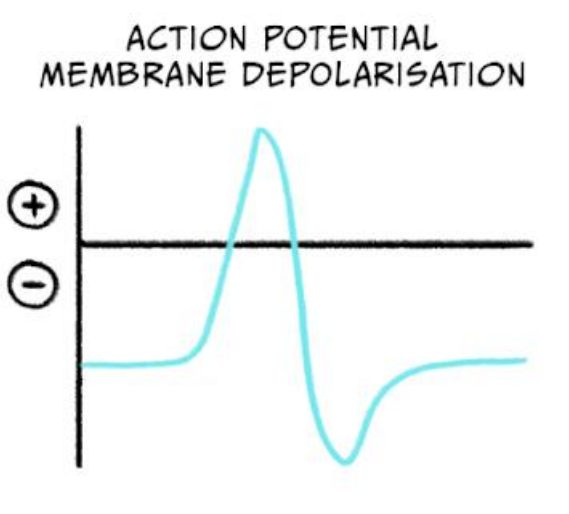
- AI is more similar to:
  - implicit memory systems (procedural memory, unconscious)
  - fast pattern completion (e.g., visual cortex)
  - associative semantic networks.

- But AI lacks:
  - intentionality
  - goals
  - subjective awareness
  - metacognition
  - autobiographical memory

    ...for the moment



The neuron



A neuron does nothing unless the collective influence of all its inputs reaches a **threshold** level.

Whenever this happens, the neuron is said to **fire**. Because a neuron either fires or does nothing, it is said to be an **all-or-none** device.

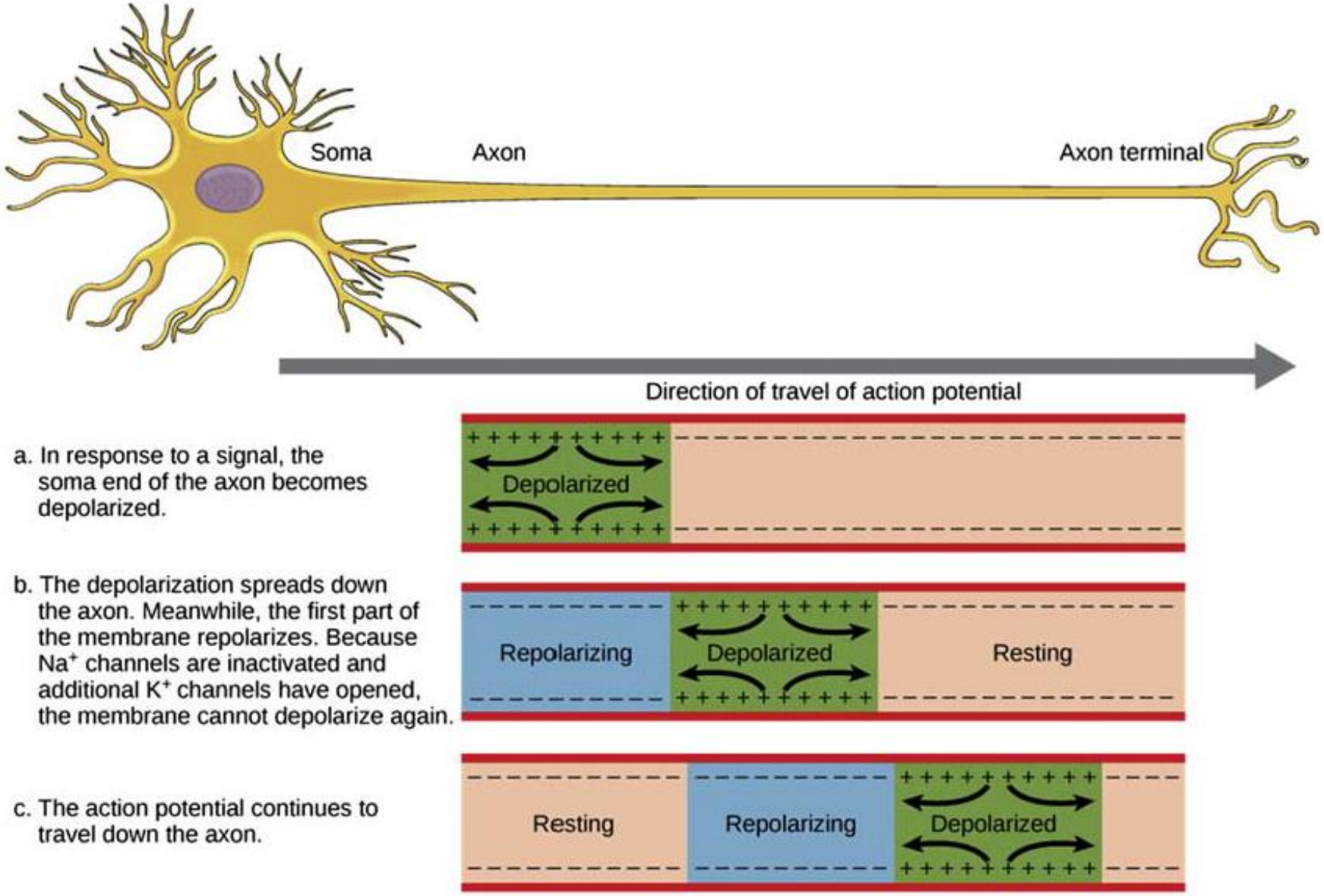
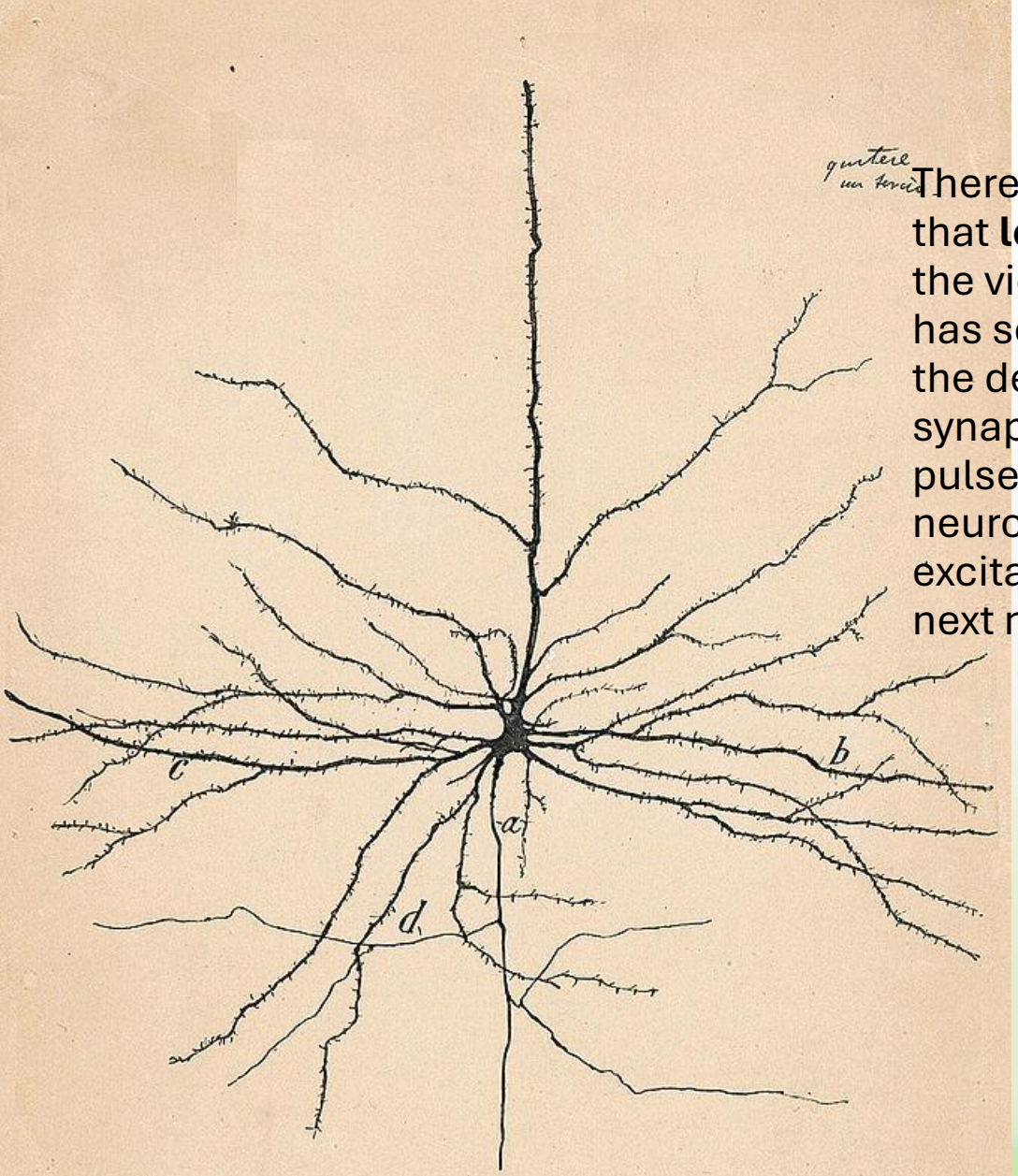
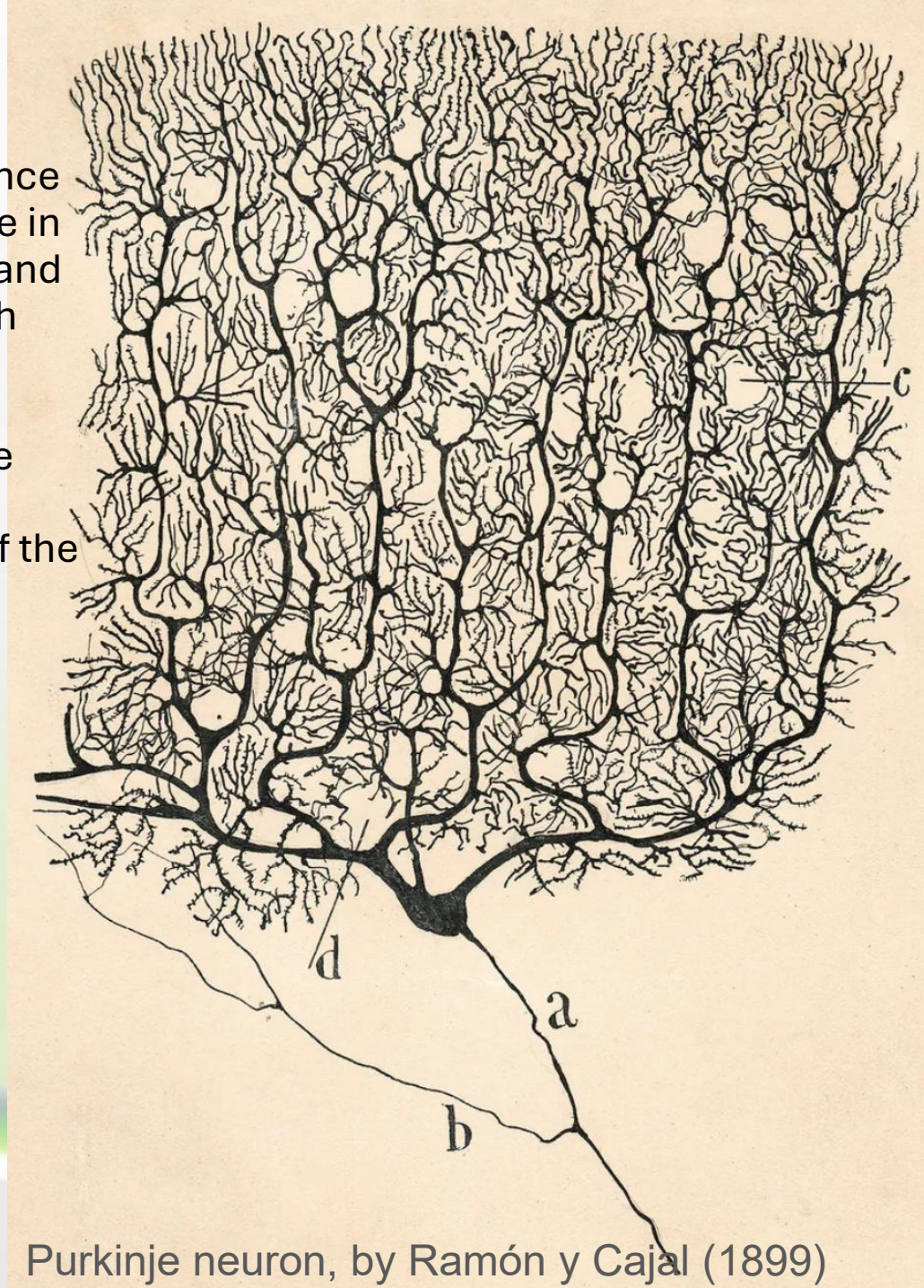


FIGURE 2.15 Action potential propagation. When the action potential occurs, the depolarization spreads down the axon towards the axon terminals where the synapse will occur. This action potential is propagated to the next neuron(s) via the synapse, thus forming the basic communication method for neurons. Note that the action potential is an *all-or-nothing event*: the neuron either fires or it does not. *Source: Public domain. Courtesy of /. Creative Commons Attribution.*

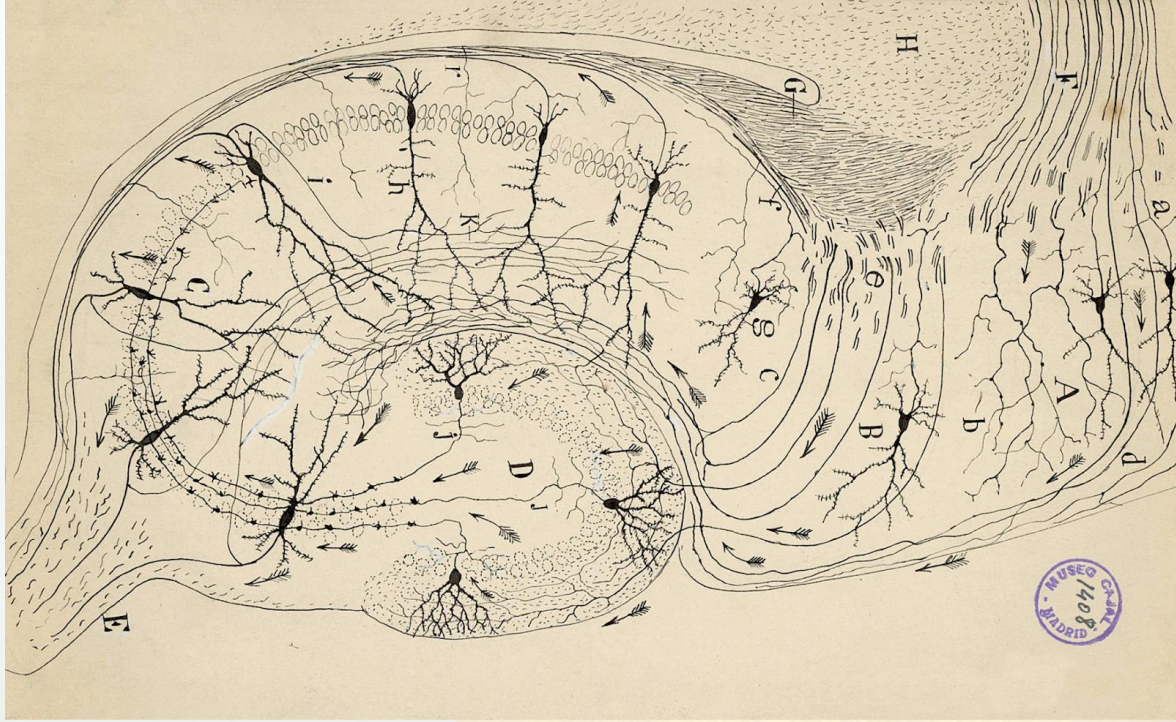
There is mounting evidence that **learning** takes place in the vicinity of synapses and has something to do with the degree to which synapses translate the pulse traveling down one neuron's axon into excitation or inhibition of the next neuron.



Pyramidal neuron, by Ramón y Cajal (1904)

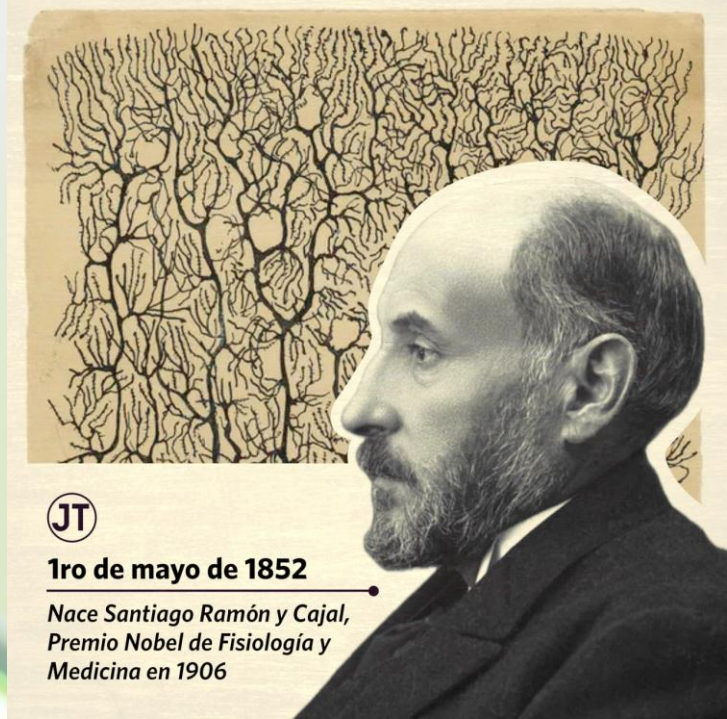


Purkinje neuron, by Ramón y Cajal (1899)



Santiago Ramón y Cajal was a neuroscience pioneer and documented many types of neurons and their connectivity.

Video: [El Legado de la Ciencia. Santiago Ramón y Cajal | Museu de les Ciències]



Neurons and Artificial neurons: Similarities

Bing Video of a neuron of a cat:

The neuron:

- **Simple cells**: **Respond to** very **specific featurese**.g., a line at a particular orientation in a particular location of the visual field.
- **Complex cells**: Respond to more **abstract features e**.g., oriented lines anywhere within a region, or movement in a direction.
- **Hierarchical organization**: The visual cortex works in layers: simple cells detect edges, complex cells detect oriented edges & movement) & higher-level areas detect shapes, objects, faces…

Neurons and Artificial neurons: Similarities

- ✔ The first layer learns **edge detectors**, like simple cells
- ✔ They respond to features like: vertical, horizontal, diagonal edges…
- ✔ As you go deeper layers respond to more **abstract patterns**
- ✔ Combine simple edges into: corners, textures, object parts, whole objects
- ✔ **hierarchical strategy**: low-level features first & high-level meaning later
- ✔ Heavily reuse the same filters across space (this is why NNs were originally seen as bio-inspired networks).
- ✔ Biological neurons connect mostly to nearby neurons → Convolutional NNs use local receptive fields.

Neurons and Artificial neurons: Differences

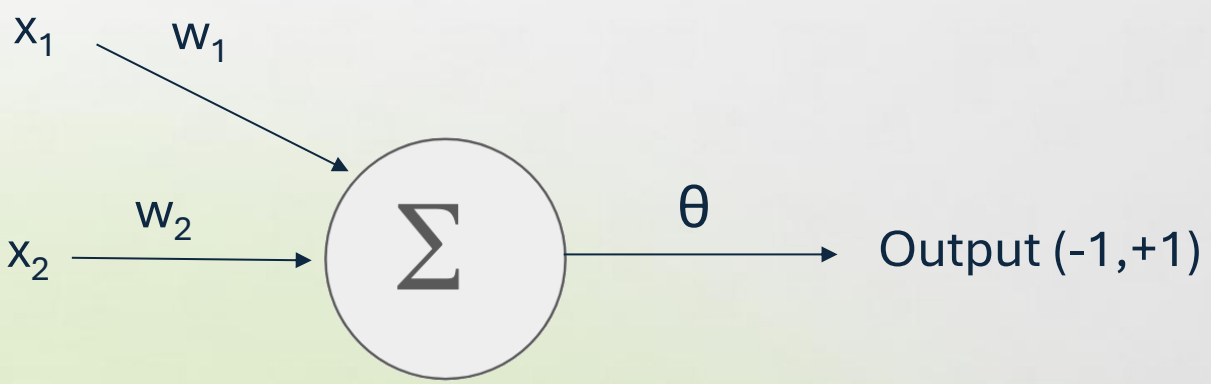
While the inspiration is strong, ANNs are not literal models of the cortex:

- ❌ neurons in the brain are far more complex
- ❌ biological vision uses massive feedback loops (NNs mostly don't)
- ❌ real neurons adapt dynamically; NN weights are fixed after training
- ❌ the cortex is nonlinear in far richer ways

Tensorflow playground

The neuron as a decision machine

- Simplest case: binary inputs, {-1, +1}
- Inputs are scaled according to "synaptic strength" and summed up
- Neurons "fires" if total sum is above a certain threshold

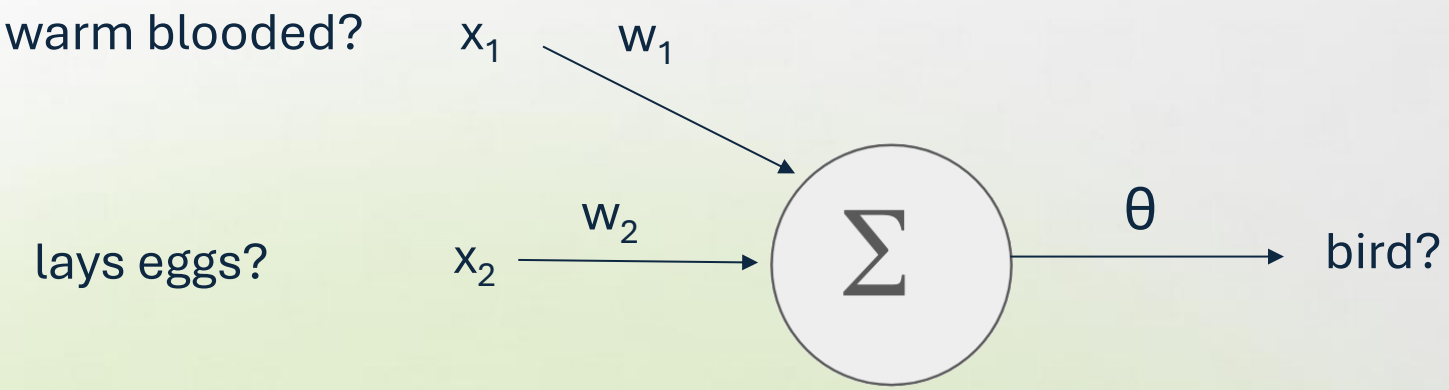


The neuron as a decision machine

Example: classifying animals into birds or not

eagle	camel	snake
+1	+1	-1
+1	-1	+1

some inputs



Q: how to implement this?
(i.e., find reasonable synaptic weights w_1, w_2, and a firing threshold θ)

Validating your classifier

- Make sure that it works on the training examples:

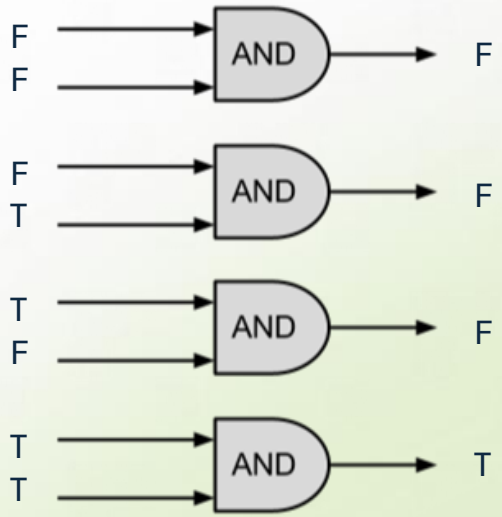
eagle	camel	snake
+1	+1	-1
+1	-1	+1

- Does it work on all possible examples?

Logical AND operator

Our first neuron tested if an animal was warm blooded **and** layed eggs.

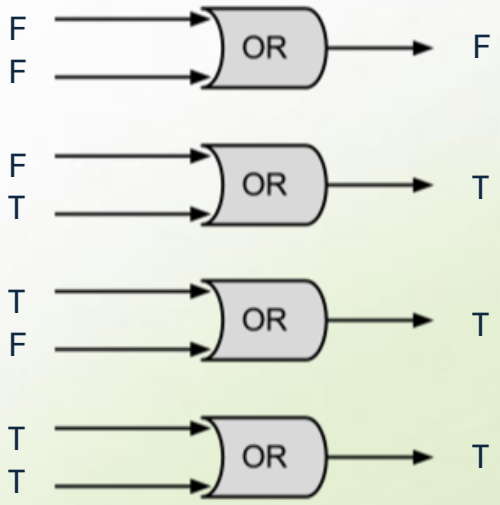
It would output +1 (true) if, and only if, both conditions were satisfied (true).
(If either was false, it would output false)



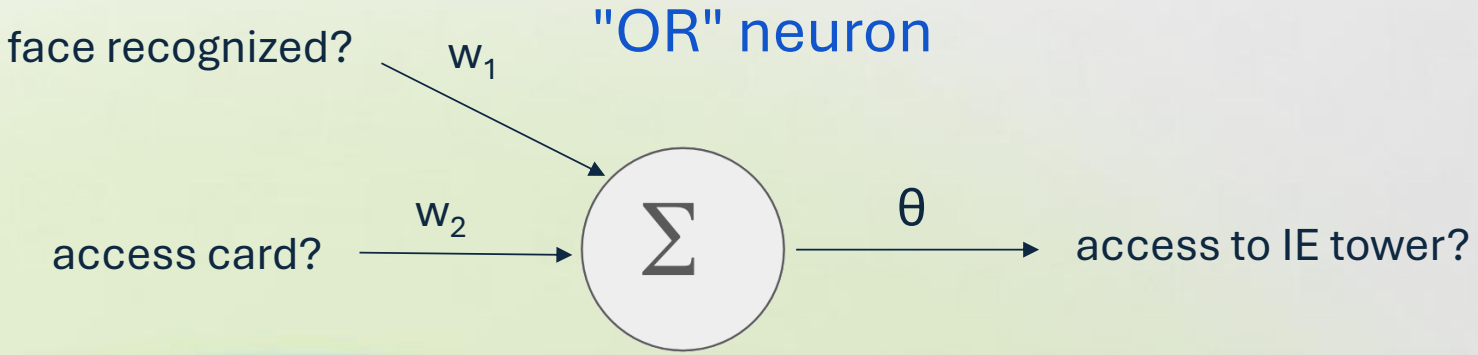
Logical OR

Can we come up with a neuron that implements the OR operator?

It should output +1 (true) if any condition is satisfied (true).
(Only outputs false if all inputs are false.)



Example scenario:



Q: how to implement the OR neuron? (find w_1, w_2, θ)

A more convenient notation

Represent inputs as: $x_1, x_2, \ldots, x_n$ with corresp. weights: $w_1, w_2, \ldots, w_n$
Represent output as y
Treat the threshold as an additional weight w_0:

Instead of checking:

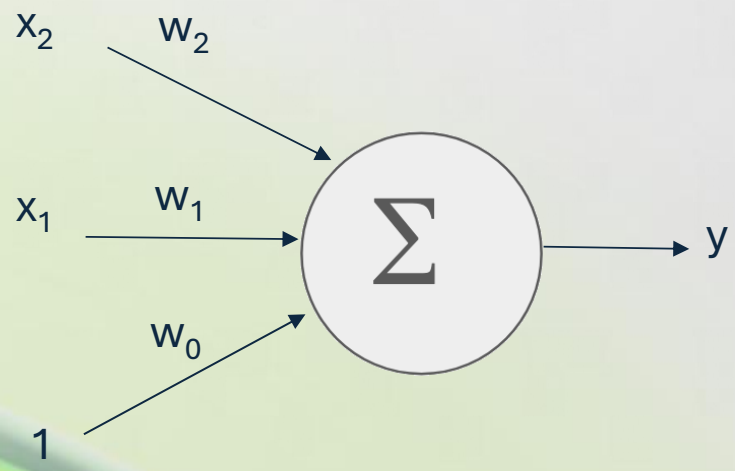
$$w1x1 + w2x2 + \ldots + wnxn > \theta$$

we rearrange it as:

$$w1x1 + w2x2 + \ldots + wnxn - \theta > 0$$

By setting w0 = -θ, we have:

$$w0 + w1x1 + w2x2 + \ldots + wnxn > 0$$



Can we solve more complicated tests?

- Can we fix the bird detecting neuron?
- E.g.:
  (warm blooded **and** lays eggs) **and** (does **not** lactate)

Q: implement the neuron above. *Hint*: inhibitory connections!

Can we solve more complicated tests?

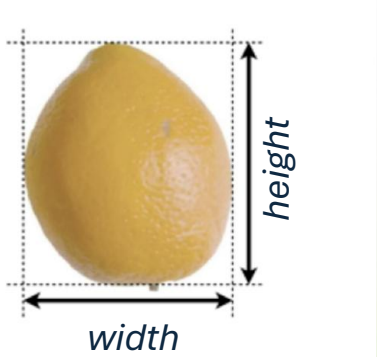
- It is possible to combine multiple logical operators into a single neuron!
- E.g., you can enter the gym if:
  (membership is active **and** you bring your id)
  or
  (you purchase a day pass **and** sign the waiver)

Q: implement the neuron above

What about non-binary inputs?

Not all problems are logical – numerical data are everywhere!

Example 1: classifying lemons in a juice factory



Source: matthewsmarking.com

	width (cm)	height (cm)
not ripe	5.6	7.2
not ripe	5.2	7.0
not ripe	5.5	7.6
ripe	5.7	7.5
ripe	5.8	7.7
ripe	5.9	7.9
ripe	6.1	8.1
ripe	6.2	8.2

Can we still use a neuron in this case?

Credit: Zemel, Urtasum, Fidler (UofT)

What about non-binary inputs?

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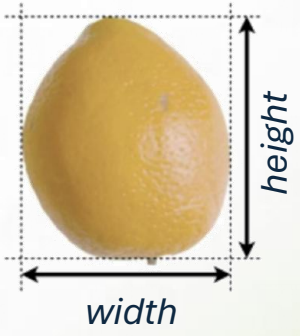
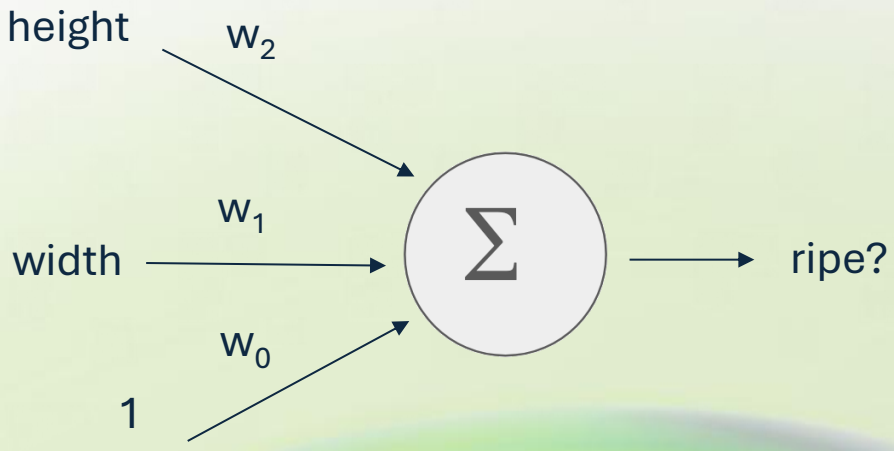
Example 1: classifying lemons





	width (cm)	height (cm)
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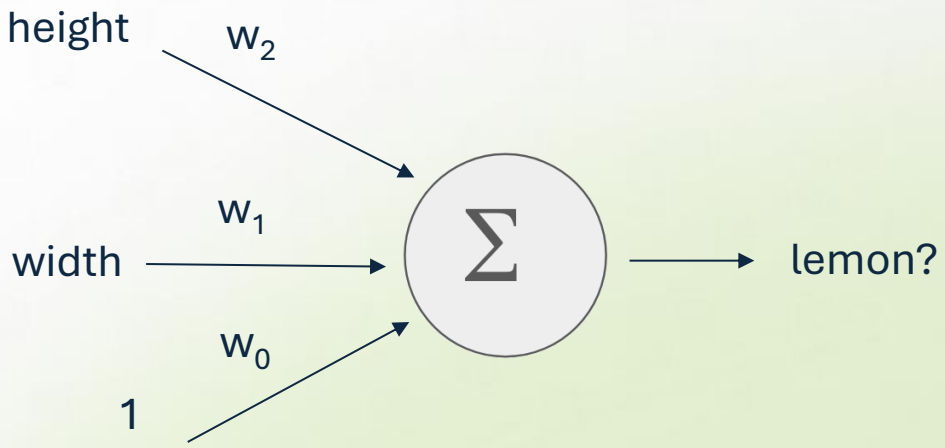
	5.2	7.0
	5.5	7.6
ripe	5.7	7.5
	5.8	7.7
	5.9	7.9
	6.1	8.1
	6.2	8.2

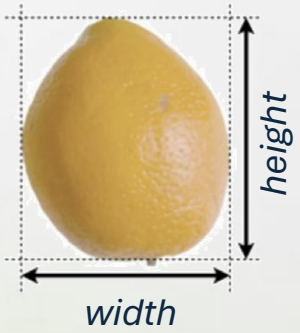
Hint: if you don't need one of the inputs, just set its corresponding weight to zero!

Q: how to classify lemons into ripe vs. not ripe? (choose w_0, w_1, w_2)

What about non-binary inputs?

Example 2: classifying fruits



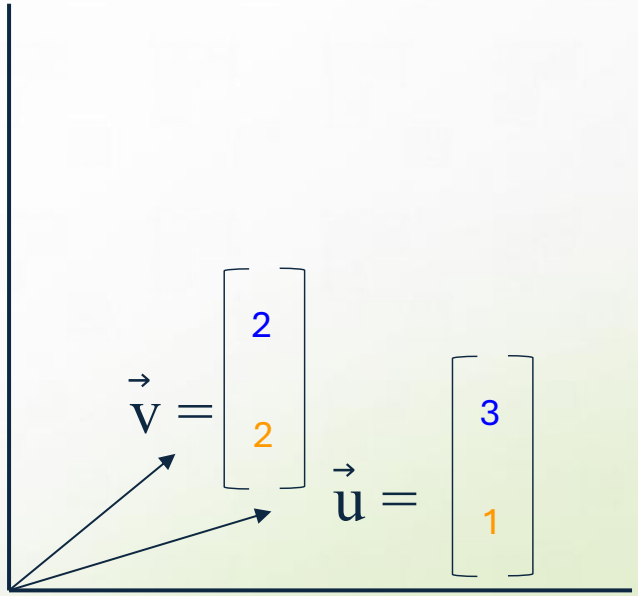


	width (cm)	height (cm)
lemons	5.7	7.2
	5.9	7.7
	6.1	8.8
	6.5	7.8
	7.5	8.3
oranges	6.9	6.5
	7.2	7.2
	7.5	7.4
	8.1	8.0
	6.3	6.2

Quiz: How to distinguish between lemons and oranges?

No longer trivial! (Try: $w_0 = 0$, $w_1 = -1$, $w_2 = 1$)

The “dot” product



$$\vec{u} + \vec{v} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\vec{u} - \vec{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = 2 \times 3 + 2 \times 1 = 6 + 2 = 8$$

Adds the products of corresponding components of two vectors…

— it produces a **scalar**!

The dot product – properties

1. $u \cdot v = v \cdot u$ (u, v, w are vectors)
2. $u \cdot (v + w) = u \cdot v + u \cdot w$
3. $(Cv) \cdot w = C\,(v \cdot w) = v \cdot (Cw)$ (C is a scalar)
4. $\vec{0} \cdot v = \vec{0}$ (all-zeroes vector)
5. $v \cdot v = ||v||^2$ (magnitude squared) Since: $||v|| = \sqrt{v_1^2 + v_2^2 + \ldots + v_n^2} = \sqrt{v \cdot v}$

Q: check for yourself that all of these properties really hold

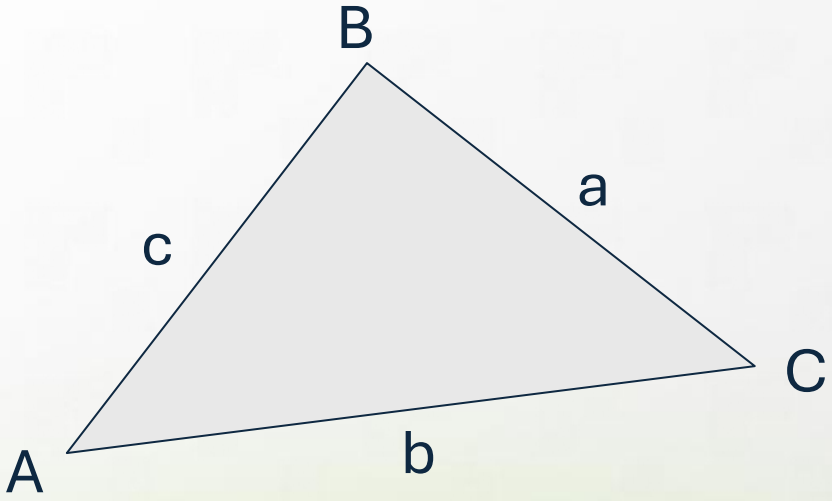
The dot product – geometric interpretation

Remember our long lost friend: the law of cosines!

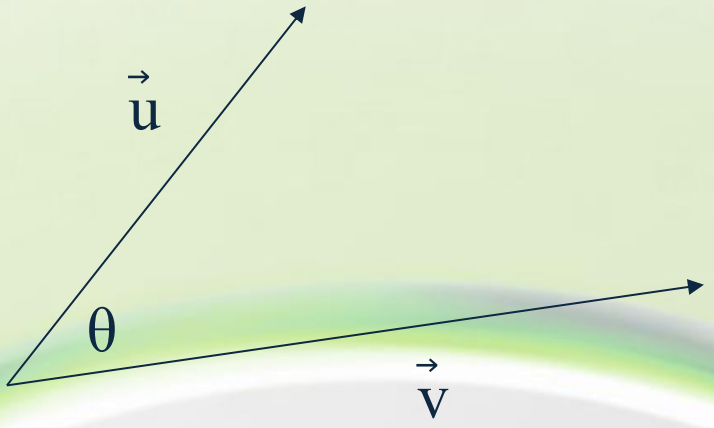
The dot product – geometric interpretation

Law of cosines:

$$a^2 = b^2 + c^2 - 2bc \cos A$$



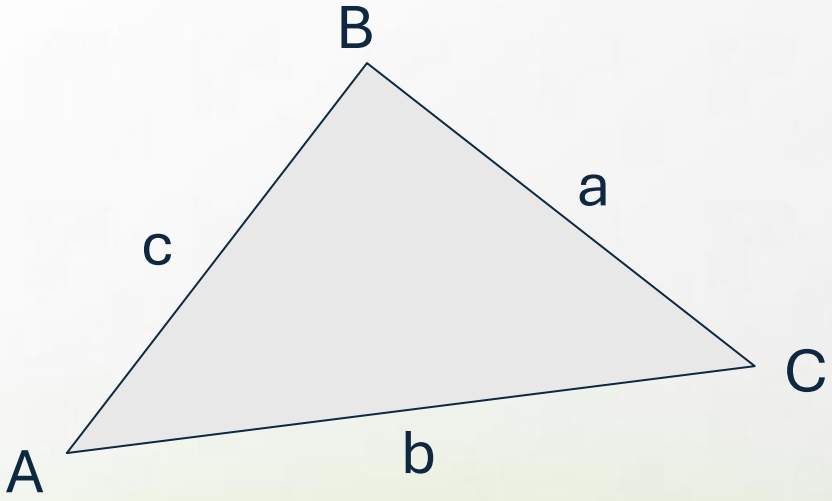
What if instead of a triangle we have two vectors?

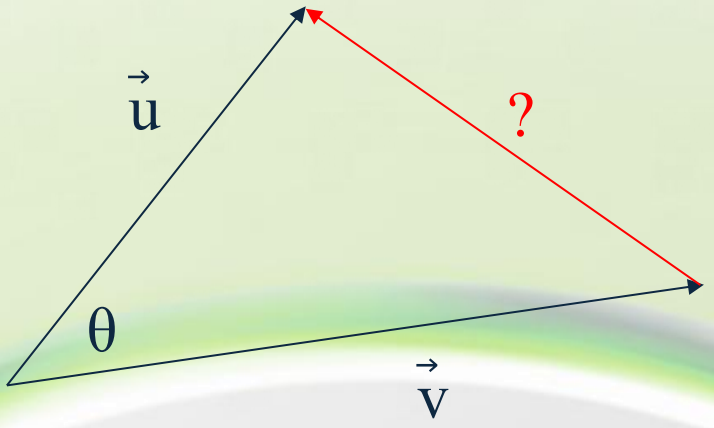


The dot product – geometric interpretation

Law of cosines:

$$a^2 = b^2 + c^2 - 2bc \cos A$$





The dot product – geometric interpretation

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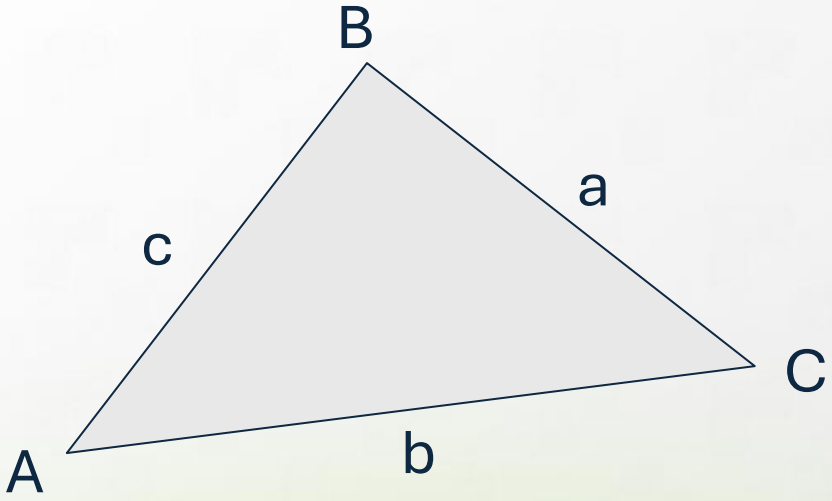
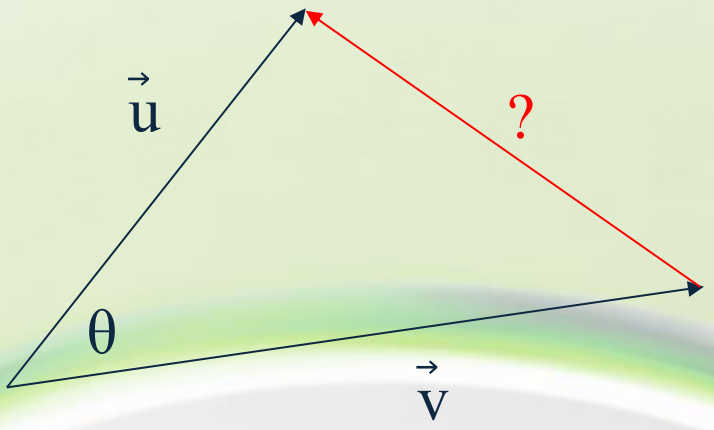




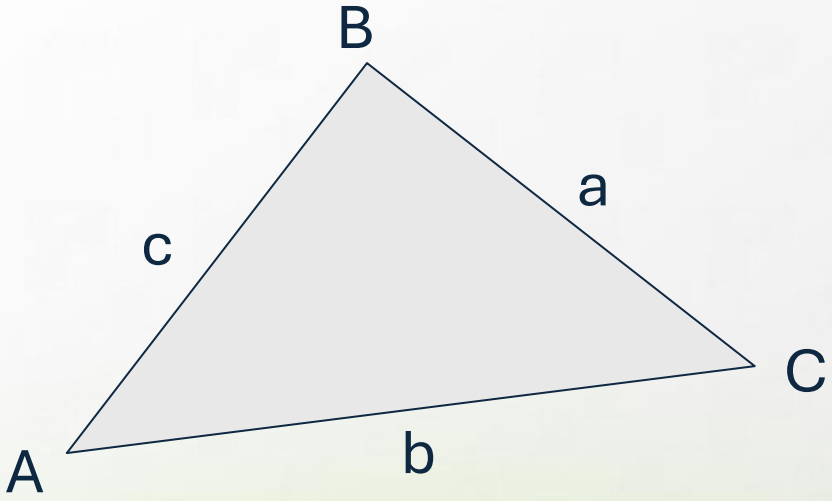
$\vec{v} + \vec{?} = \vec{u}$

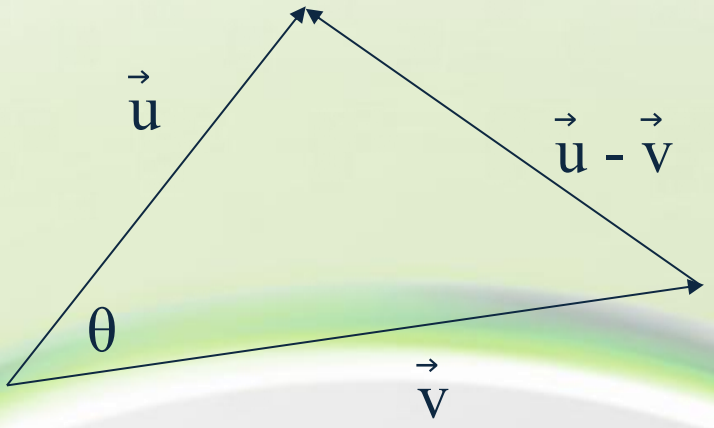
$\vec{?} =$

The dot product – geometric interpretation

Law of cosines:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

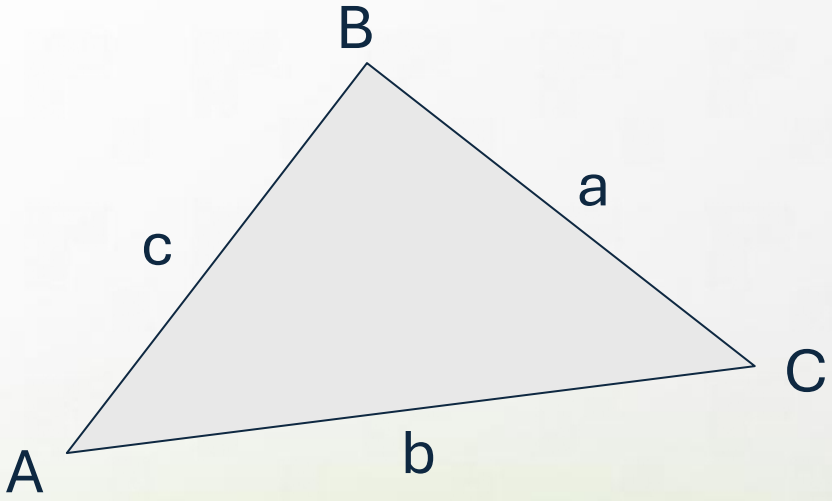


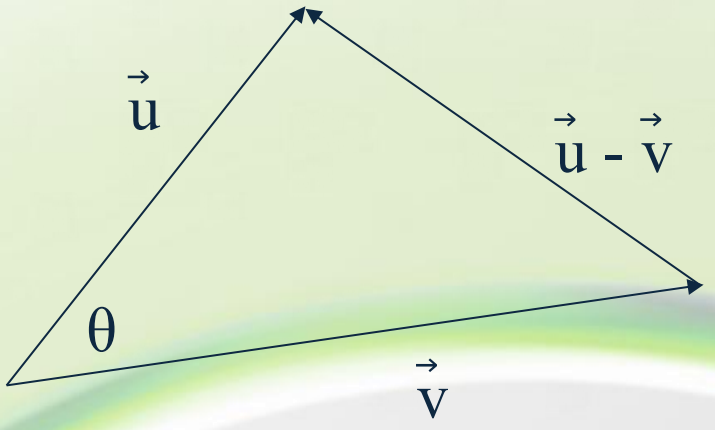


The dot product – geometric interpretation

Law of cosines:

$$a^2 = b^2 + c^2 - 2bc \cos A$$





Now apply the identity:

$$\|u - v\|^2 = \|v\|^2 + \|u\|^2 - 2\|v\|\,\|u\|\cos\theta$$

$$(u-v)\cdot(u-v) =$$

$$u\cdot u - 2v\cdot u + v\cdot v =$$

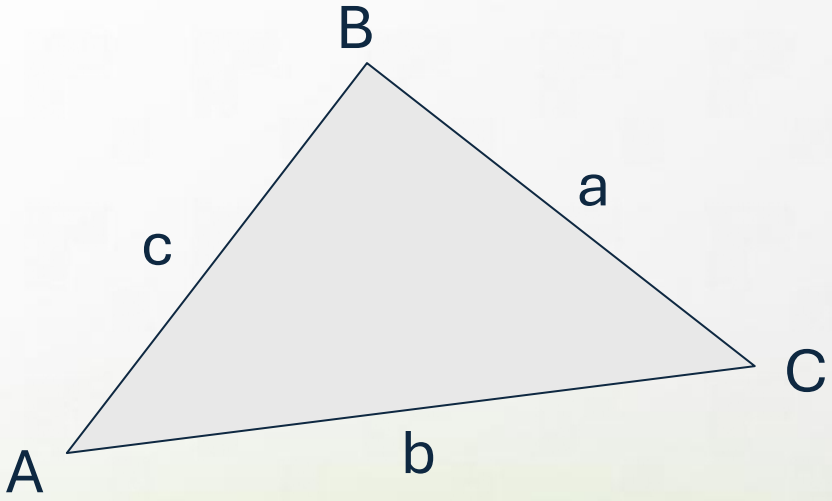
$$\cancel{\|u\|^2} - 2v\cdot u + \cancel{\|v\|^2} = \cancel{\|v\|^2} + \cancel{\|u\|^2} - 2\|v\|\,\|u\|\cos\theta$$

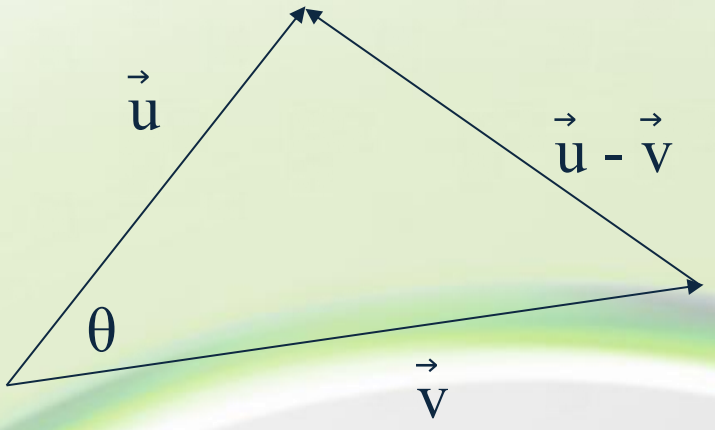
$$-2v\cdot u = -2\|v\|\,\|u\|\cos\theta$$

The dot product – geometric interpretation

Law of cosines:

$$a^2 = b^2 + c^2 - 2bc \cos A$$





Now apply the identity:

$$\|u - v\|^2 = \|v\|^2 + \|u\|^2 - 2\|v\|\,\|u\|\cos\theta$$

$$(u-v)\cdot(u-v) =$$

$$u\cdot u - 2v\cdot u + v\cdot v =$$

$$\cancel{\|u\|^2} - 2v\cdot u + \cancel{\|v\|^2} = \cancel{\|v\|^2} + \cancel{\|u\|^2} - 2\|v\|\,\|u\|\cos\theta$$

$$\cancel{-}\,2v\cdot u = \cancel{-}\,2\|v\|\,\|u\|\cos\theta$$

$$v\cdot u = \|v\|\,\|u\|\cos\theta$$

The dot product – geometric interpretation

(Important) conclusions:

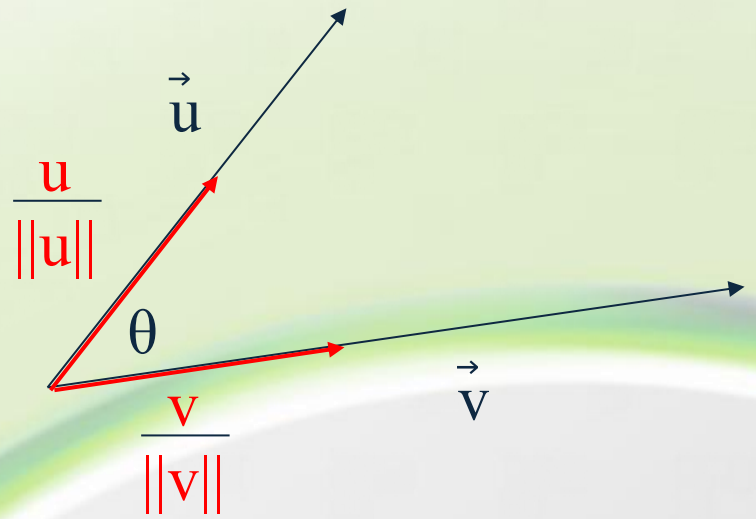
$$v \cdot u = \|v\| \, \|u\| \cos \theta$$

$$\cos \theta = \frac{v \cdot u}{\|v\| \, \|u\|}$$

Effectively computes the dot product between two unit vectors

$$\theta = \cos^{-1} v \cdot u$$

Must be a number between -1 and +1



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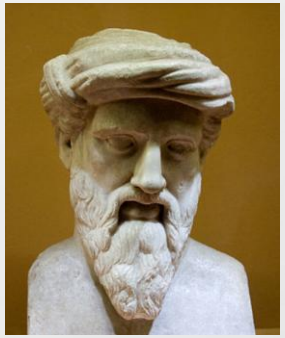
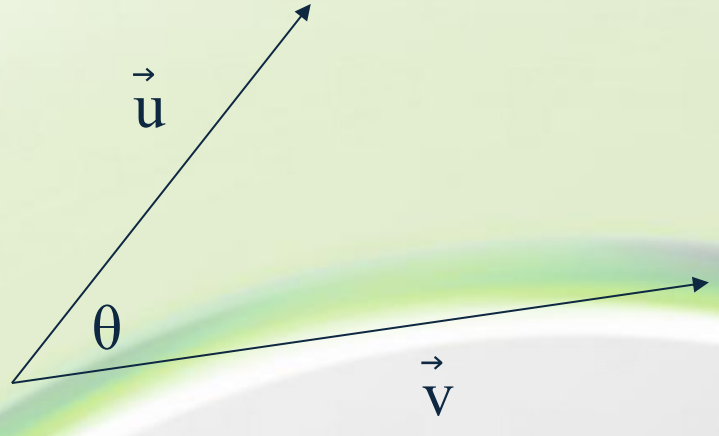


Q1: What happens if u and v are orthogonal?

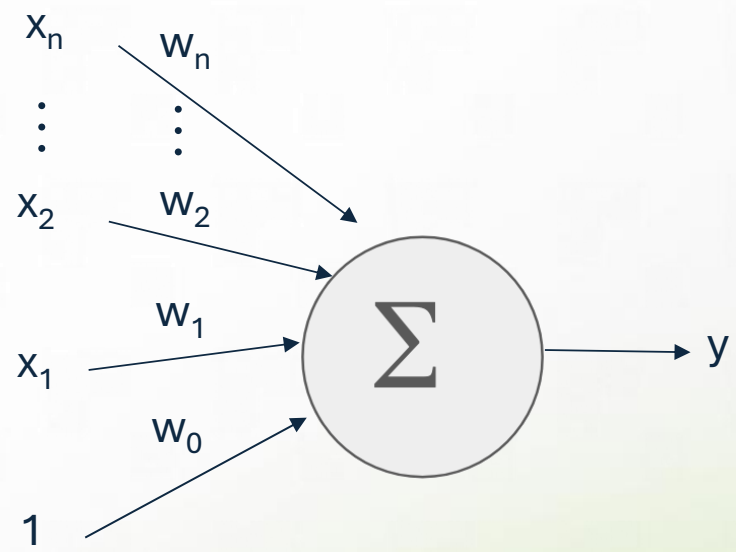
Q2: What happens if u and v point in the same direction?

What about the opposite direction? ($\theta = 180^{o}$)



What does all this have to do with neurons?

Recall:



The output of our artificial neuron was computed as:

$$y = \begin{cases} +1 \text{ if } w_0 + w_1x_1 + w_2x_2 + \ldots + w_nx_n > 0 \\ -1 \text{ otherwise.} \end{cases}$$

But:

(e.g., for n=3)

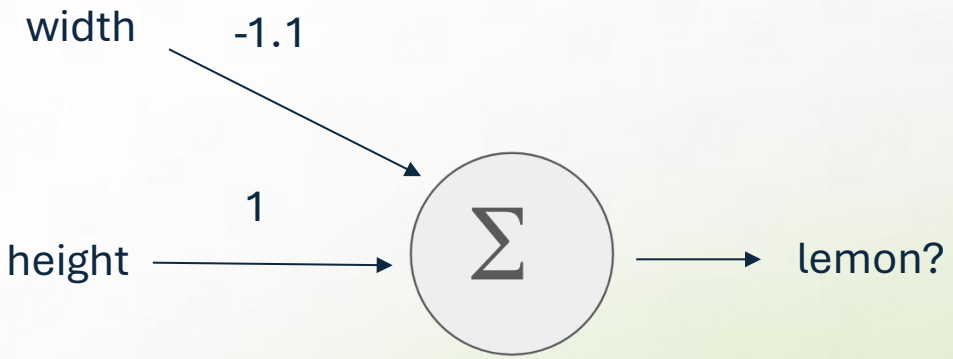
$$\begin{pmatrix} x_3 \\ x_2 \\ x_1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} w_3 \\ w_2 \\ w_1 \\ w_0 \end{pmatrix} = \vec{x} \cdot \vec{w} = w_0 + w_1x_1 + w_2x_2 + w_3x_3 = \sum_{i=0}^{n} w_ix_i$$

Can express decision rule as:

$$y = \begin{cases} +1 \text{ if } \vec{x} \cdot \vec{w} > 0 \\ -1 \text{ otherwise.} \end{cases}$$

The geometry behind a neuron's decision rule

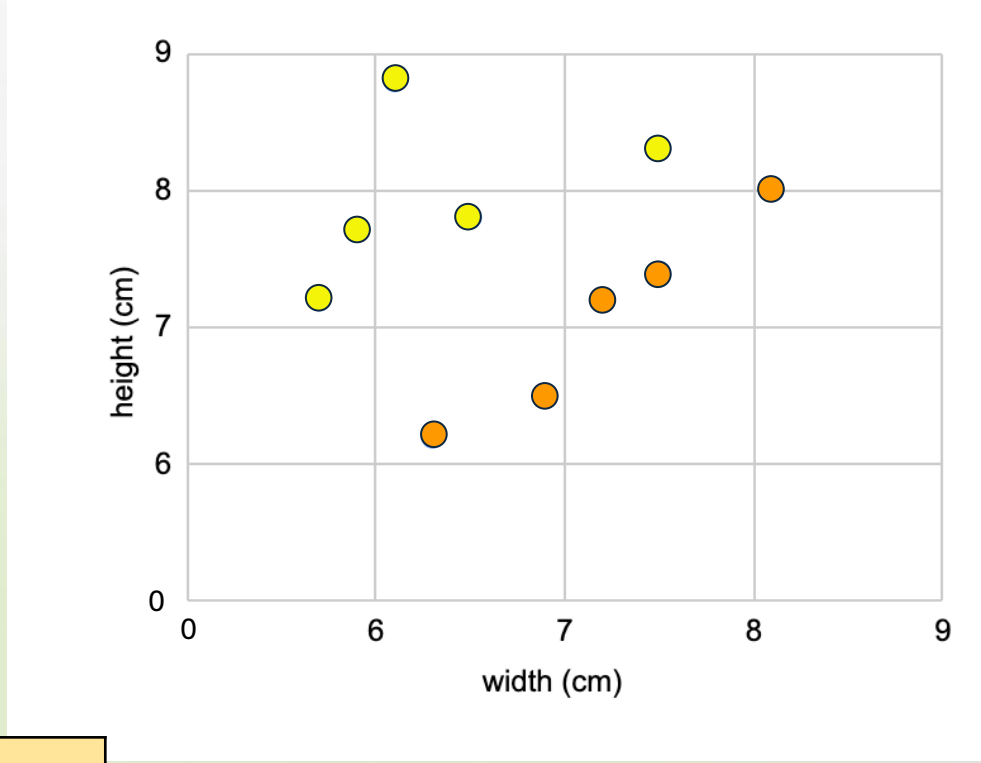
Recall the neuron below:



Example inputs:

width	5.7	5.9	6.1	6.5	7.5	6.9	7.2	7.5	8.1	8.3
height	7.2	7.7	8.8	7.8	8.3	6.5	7.2	7.4	8.0	8.1

We can plot these as points in 2-D:



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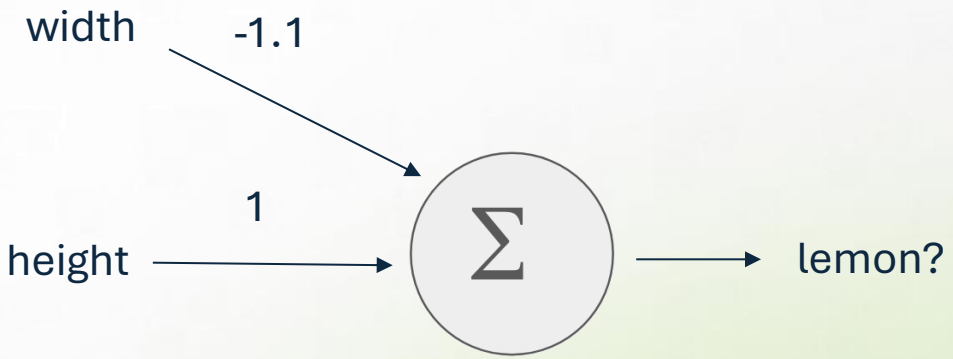
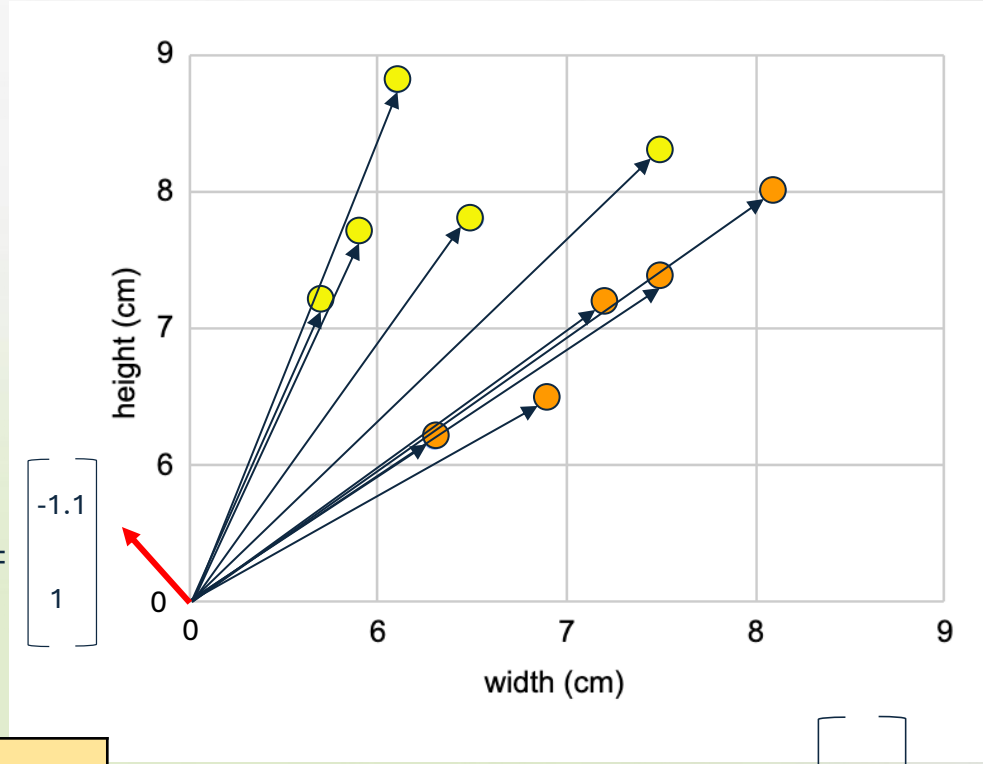


We can plot these as ~~points~~ vectors in 2-D:



$$\vec{w} = \begin{bmatrix} -1.1 \\ 1 \end{bmatrix}$$

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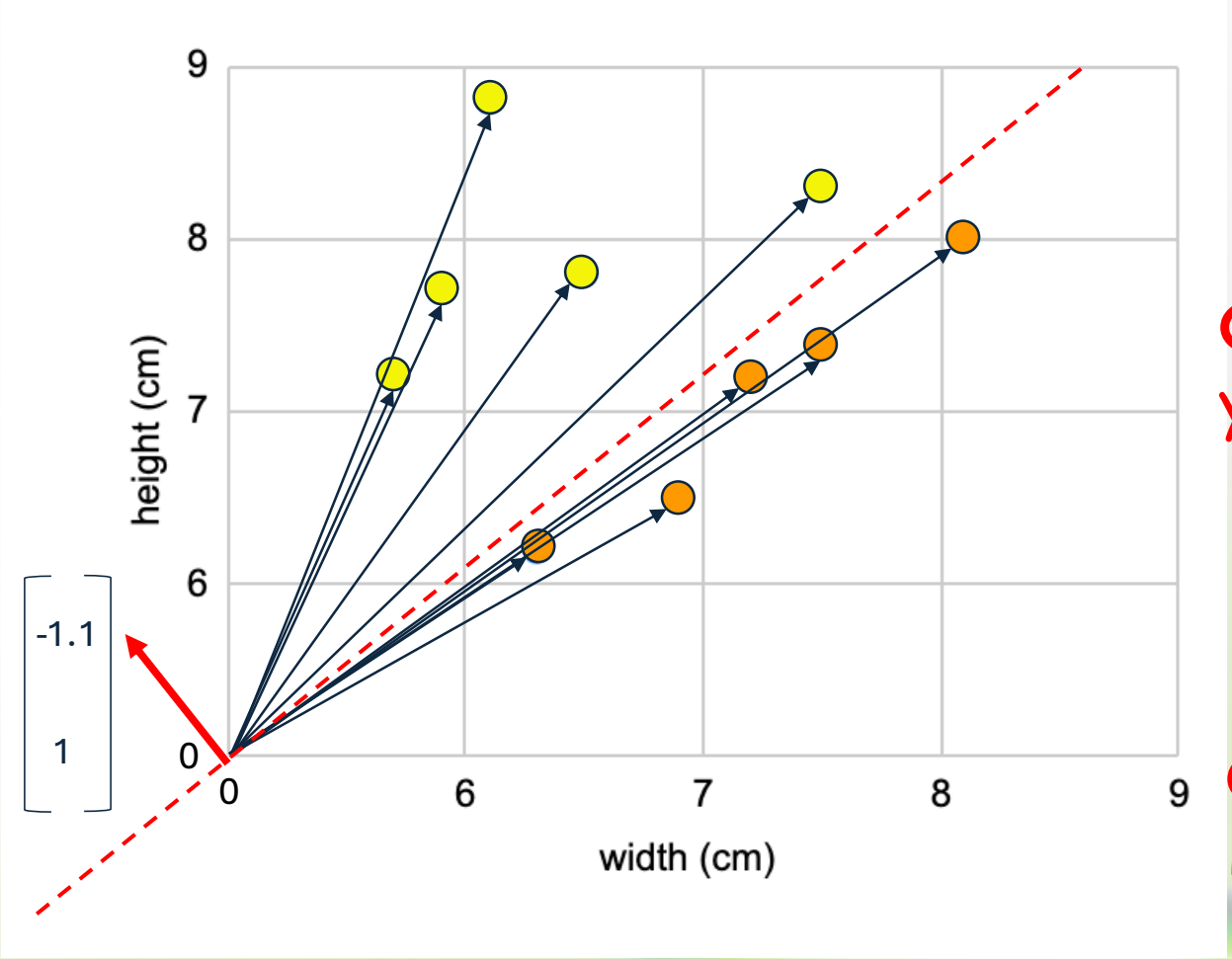
width	5.7	5.9	6.1	6.5	7.5	6.9	7.2	7.5	8.1	8.3
height	7.2	7.7	8.8	7.8	8.3	6.5	7.2	7.4	8.0	8.1

What about the weights?

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

The geometry behind a neuron's decision rule

$$\vec{w} = \begin{bmatrix} -1.1 \\ 1 \end{bmatrix}$$



Recall: the neuron decision rule

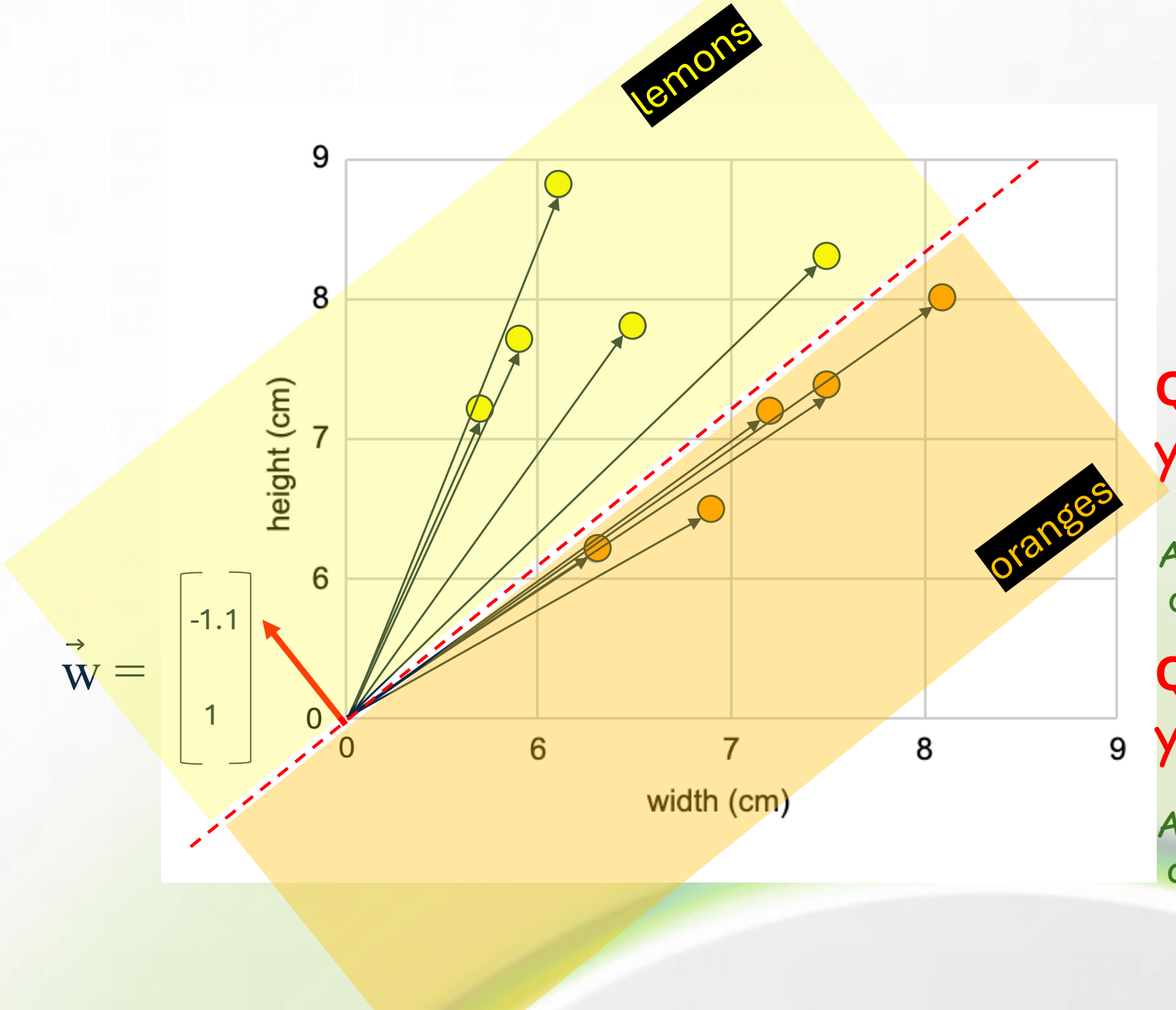
$$y = \begin{cases} +1 & \text{if } \vec{w} \cdot \vec{x} > 0 \\ -1 & \text{otherwise.} \end{cases}$$

Q: Which vectors x would yield $\vec{w} \cdot \vec{x} = 0$?

All vectors perpendicular to w! (along the dashed line)

Q: What is the dashed line?

The geometry behind a neuron's decision rule



Recall: the neuron decision rule

$$y = \begin{cases} +1 & \text{if } \vec{w} \cdot \vec{x} > 0 \\ -1 & \text{otherwise.} \end{cases}$$

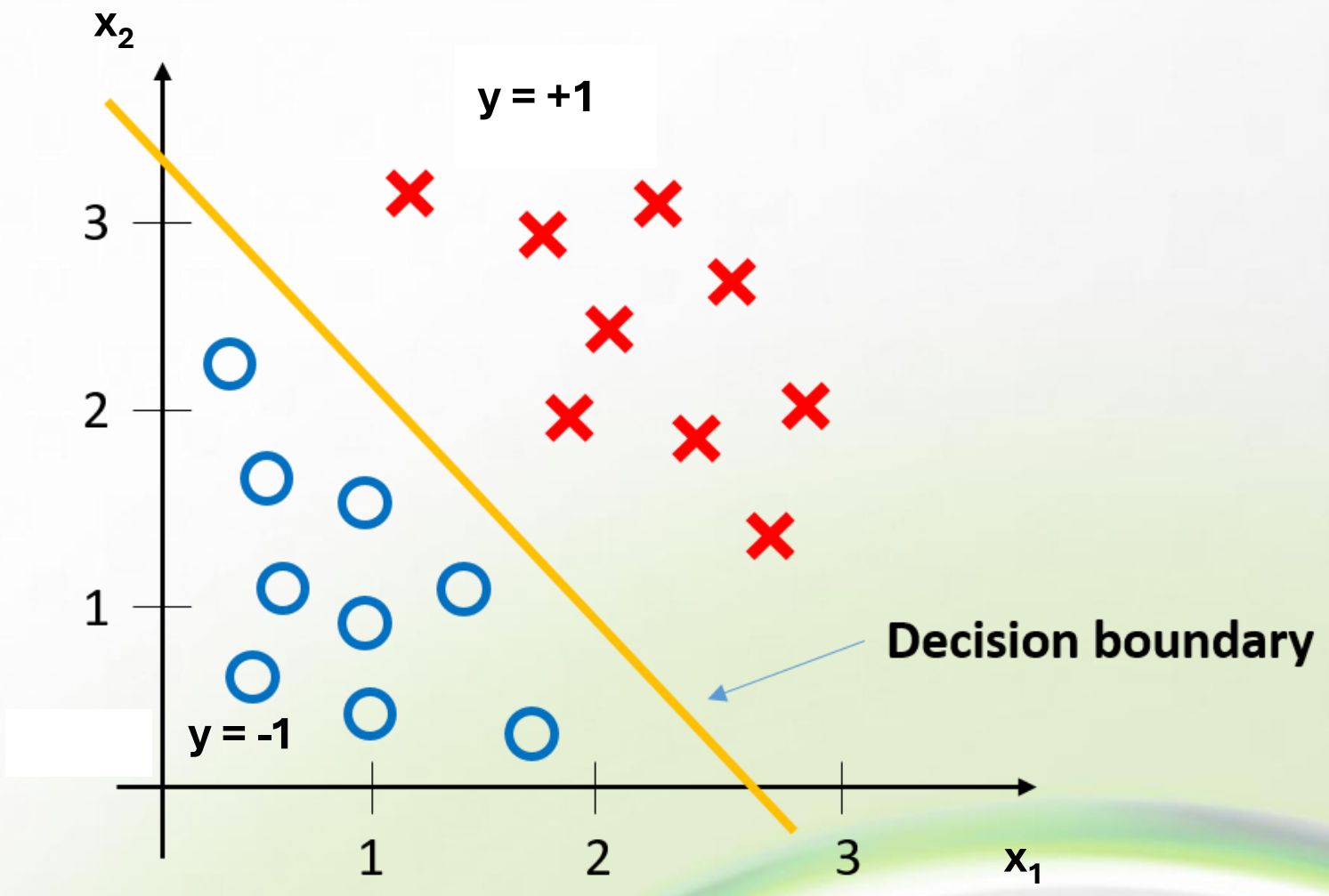
Q: Which vectors x would yield $\vec{w} \cdot \vec{x} > 0$?

All vectors on the same side of the dashed line as w (angle < 90°).

Q: Which vectors x would yield $\vec{w} \cdot \vec{x} < 0$?

All vectors on the opposite side of the dashed line as w (angle > 90°).

Q: Can you find the neuron that solve this problem?

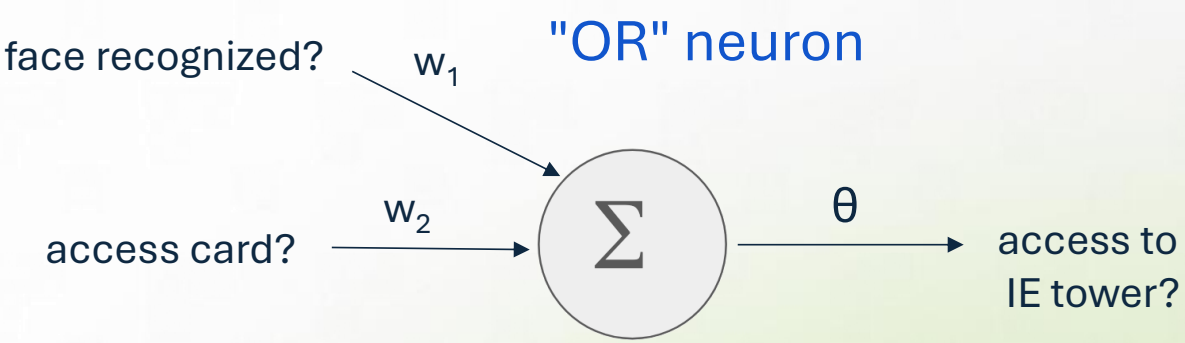


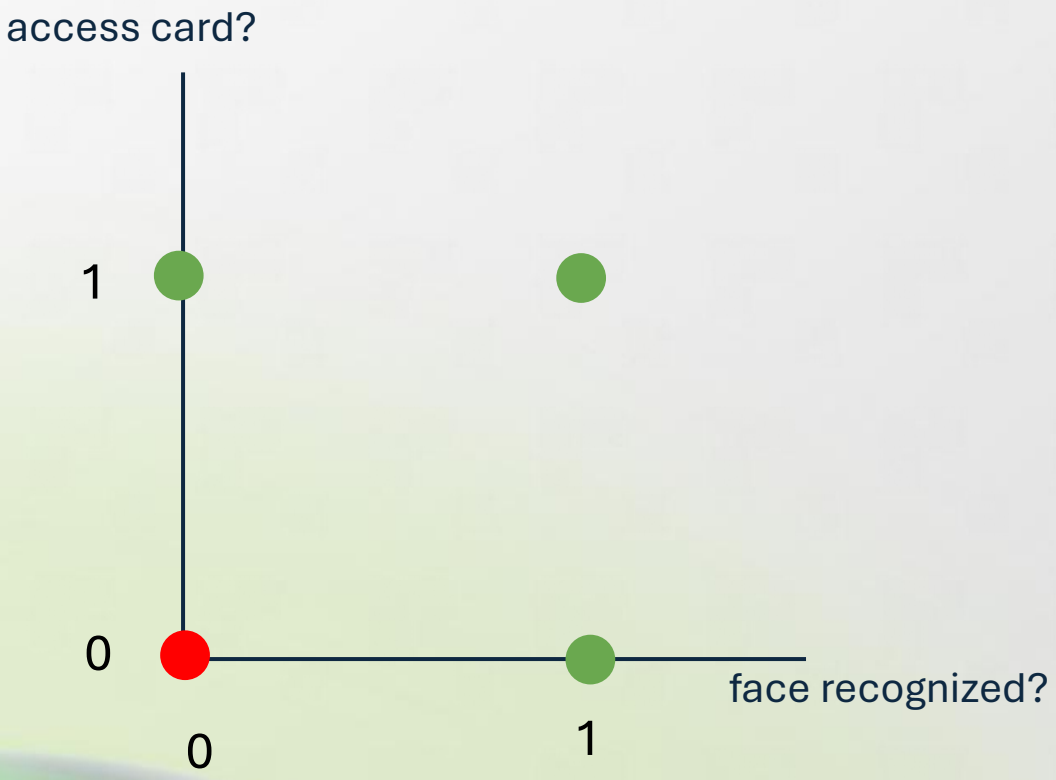
Hint: start from the line equation

Moment of truth: a neuron is nothing more than a linear inequation!

Binary problems can also be solved this way:

Recall:





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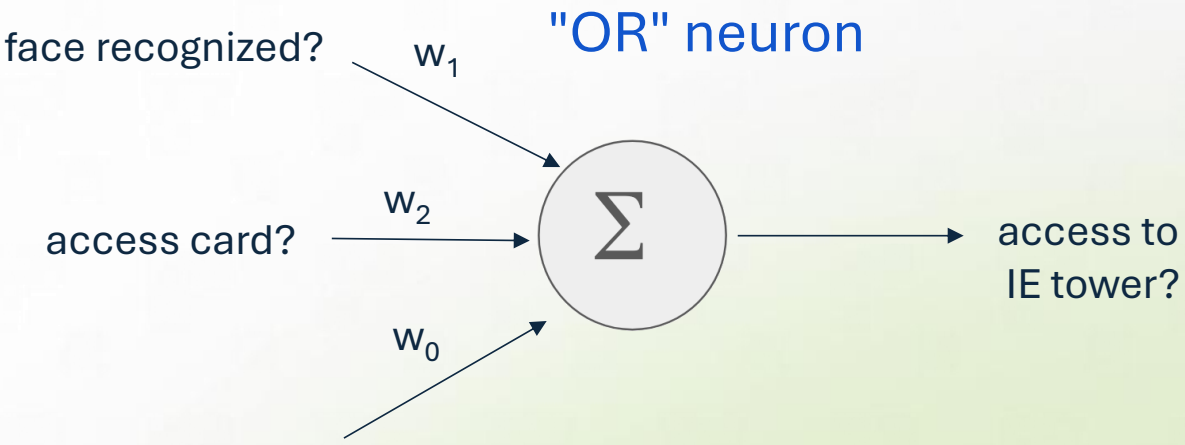
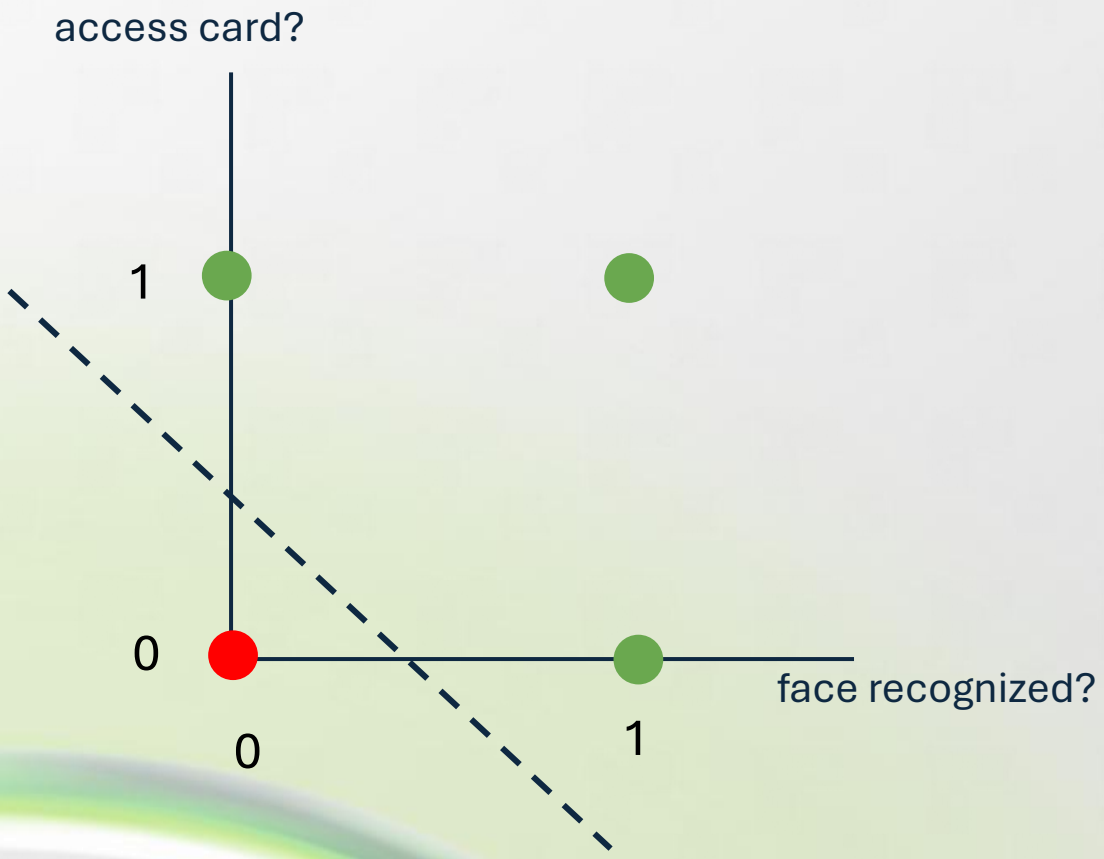
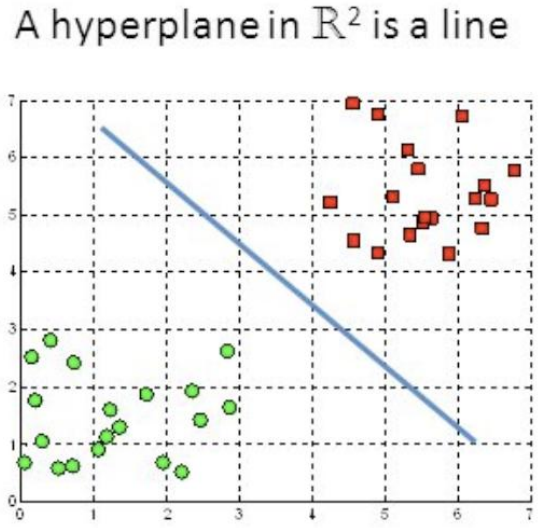




What about higher dimensions?

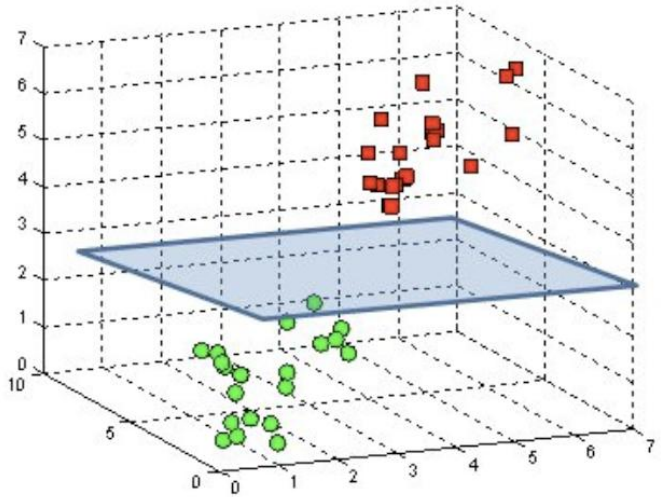
With more inputs, the boundary is still linear: point -> line -> plane -> hyperplane

A hyperplane in $\mathbf{R}^d$ is a hyperplane



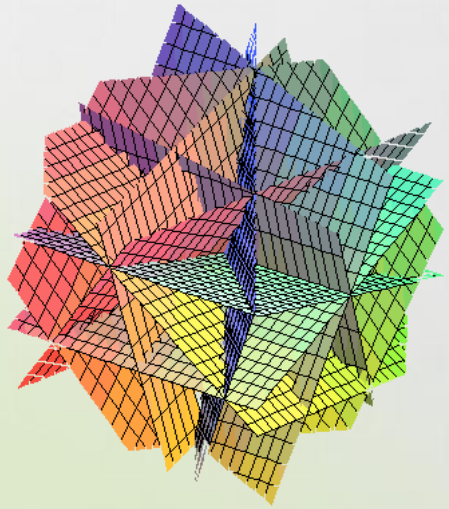
$w \cdot x = 0$

$w_0 + w_1x + w_2y = 0$



$w \cdot x = 0$

$w_0 + w_1x + w_2y + w_3z = 0$

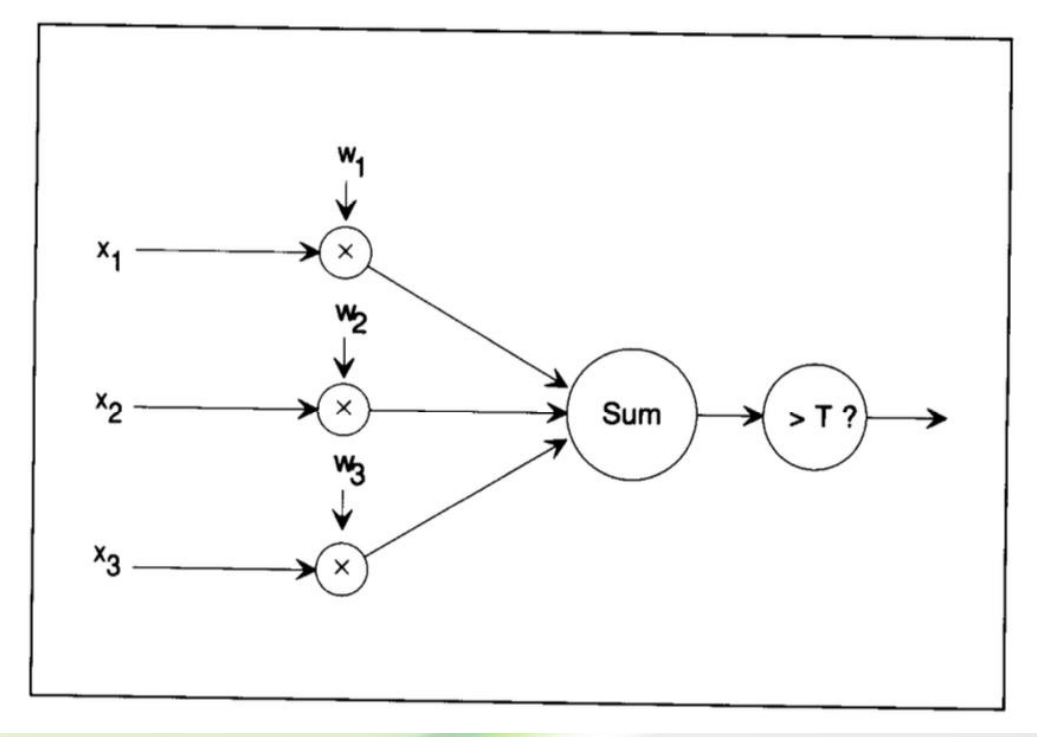


$w \cdot x = 0$

$w_0 + w_1x_1 + w_2x_2 + \ldots + w_dx_d = 0$

How to learn the decision boundary/weights automatically?

We need an algorithm for that: enter the *perceptron*



Wikipedia

Frank Rosenblatt